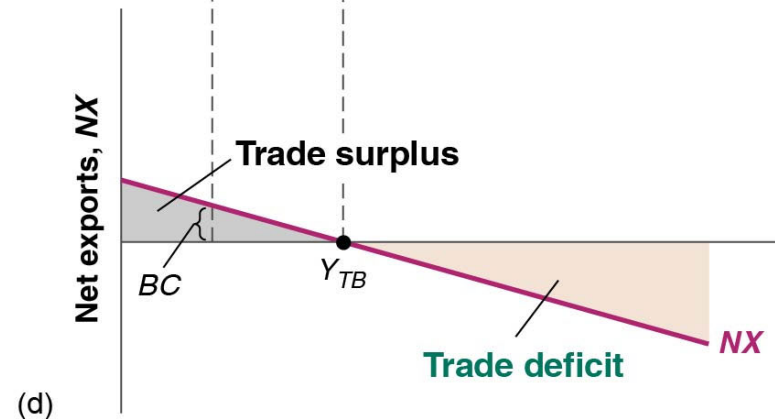
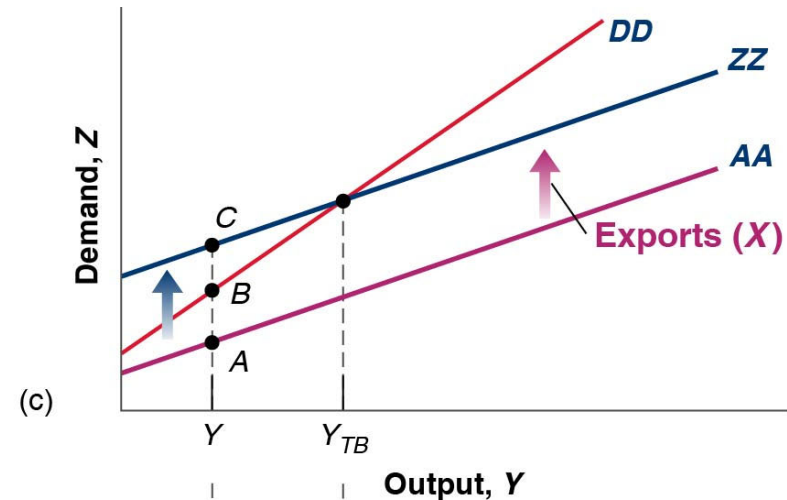
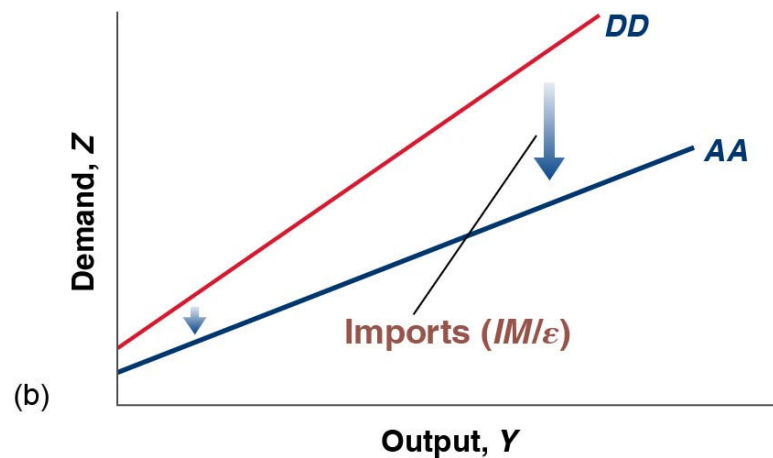
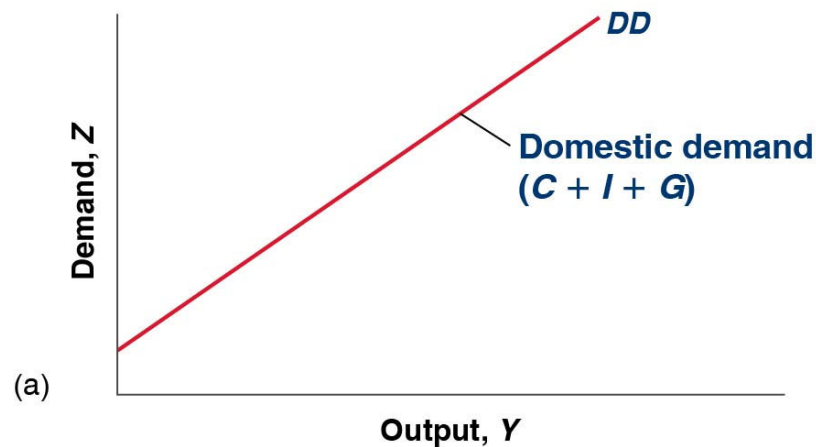
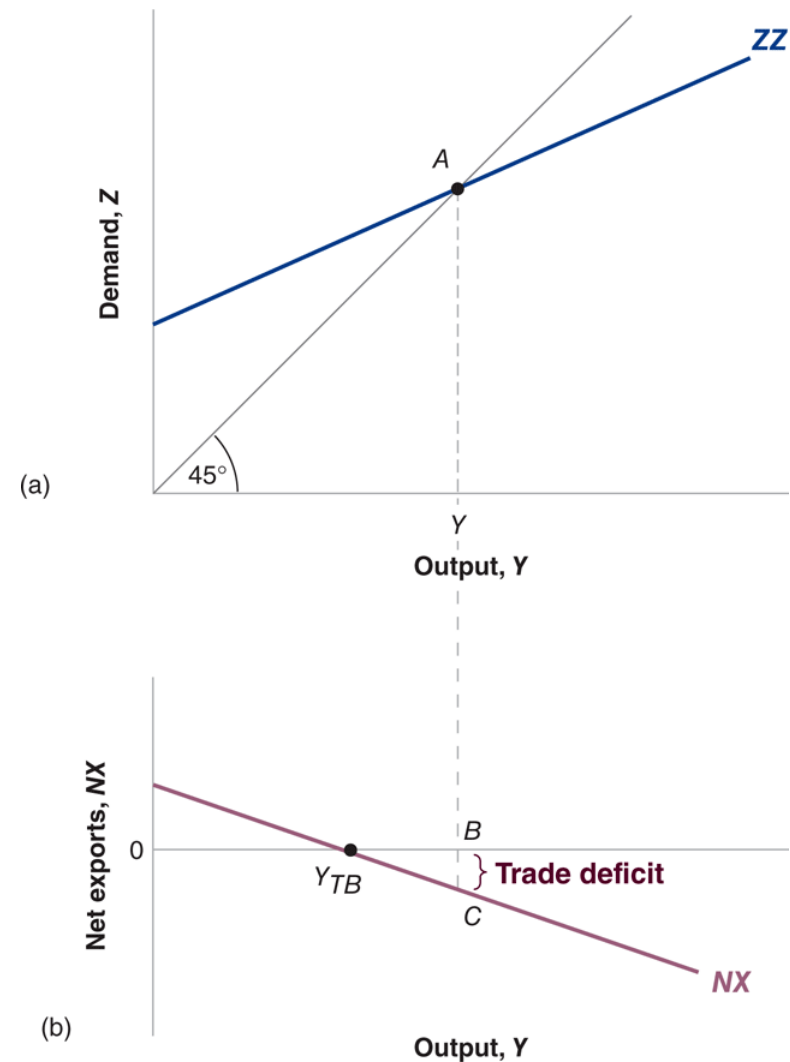


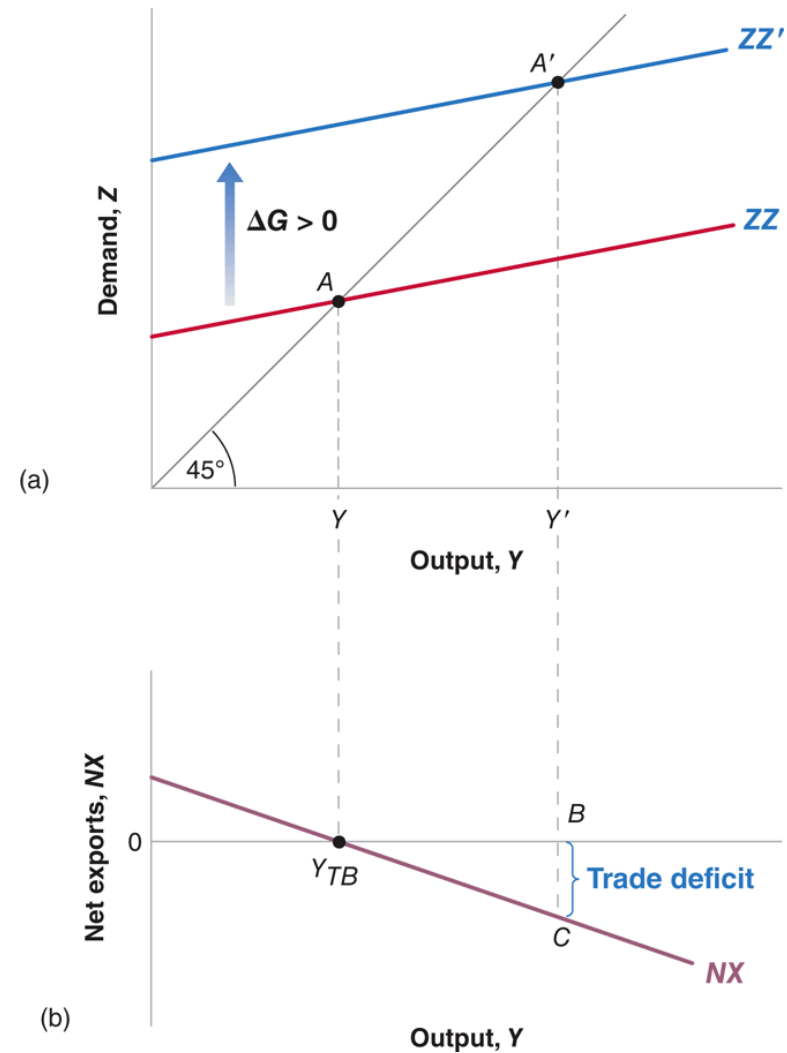
18.1 The IS Relation in the Open Economy



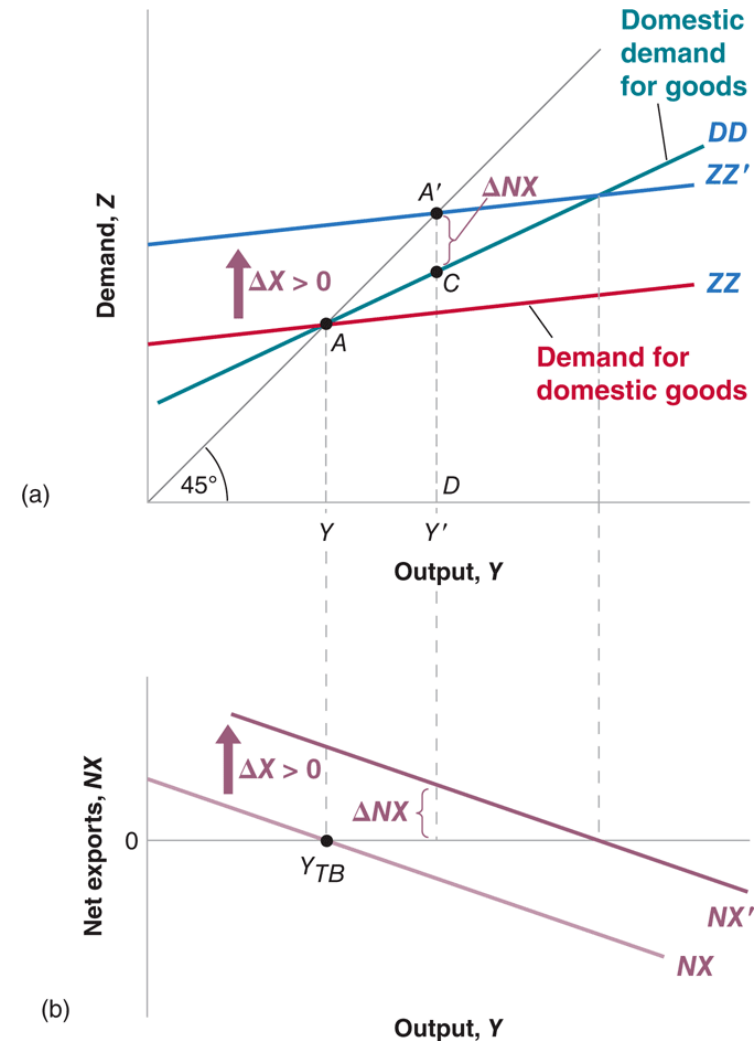
18.2 Equilibrium Output and the Trade Balance



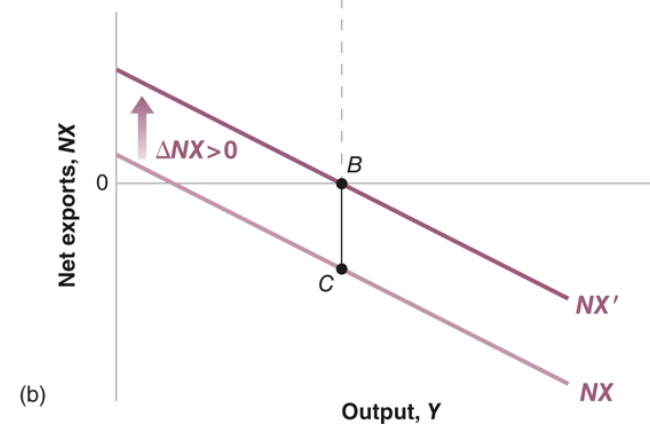
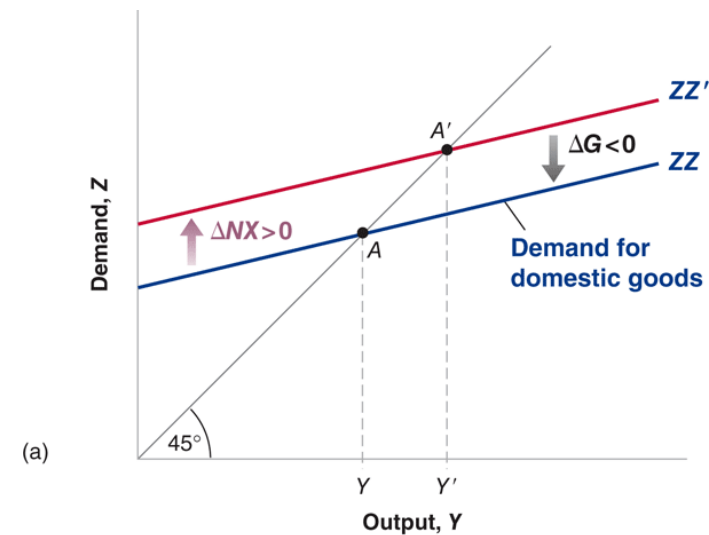
18.3 Increases in Demand–Domestic or Foreign



18.3 Increases in Demand–Domestic or Foreign



18.4 Depreciation, the Trade Balance, and Output



APPENDIX: Derivation of the Marshall-Lerner Condition

- The Marshall-Lerner condition is the condition under which a real depreciation (lower U) leads to an increase in net exports.
- Given the definition of net exports: $NX = X - IM/\varepsilon$
- Multiply both sides by ε to get: $\varepsilon NX = \varepsilon X - IM$, which can be written as: $\varepsilon(\Delta NX) = (\Delta \varepsilon)X + \varepsilon(\Delta X) - (\Delta IM)$
- Divide both sides by εX to get:

$$\frac{[\varepsilon(\Delta NX)]}{\varepsilon X} = \frac{[(\Delta \varepsilon)X]}{\varepsilon X} + \frac{[\varepsilon(\Delta X)]}{\varepsilon X} - \frac{[\Delta IM]}{\varepsilon X}$$

- If trade is initially balanced ($\varepsilon X = IM$), then:

$$\frac{(\Delta NX)}{X} = \frac{(\Delta \varepsilon)}{\varepsilon} + \frac{(\Delta X)}{X} - \frac{\Delta IM}{IM}$$